

Limits in the detection of m⁶A changes using MeRIP/m⁶A-seq

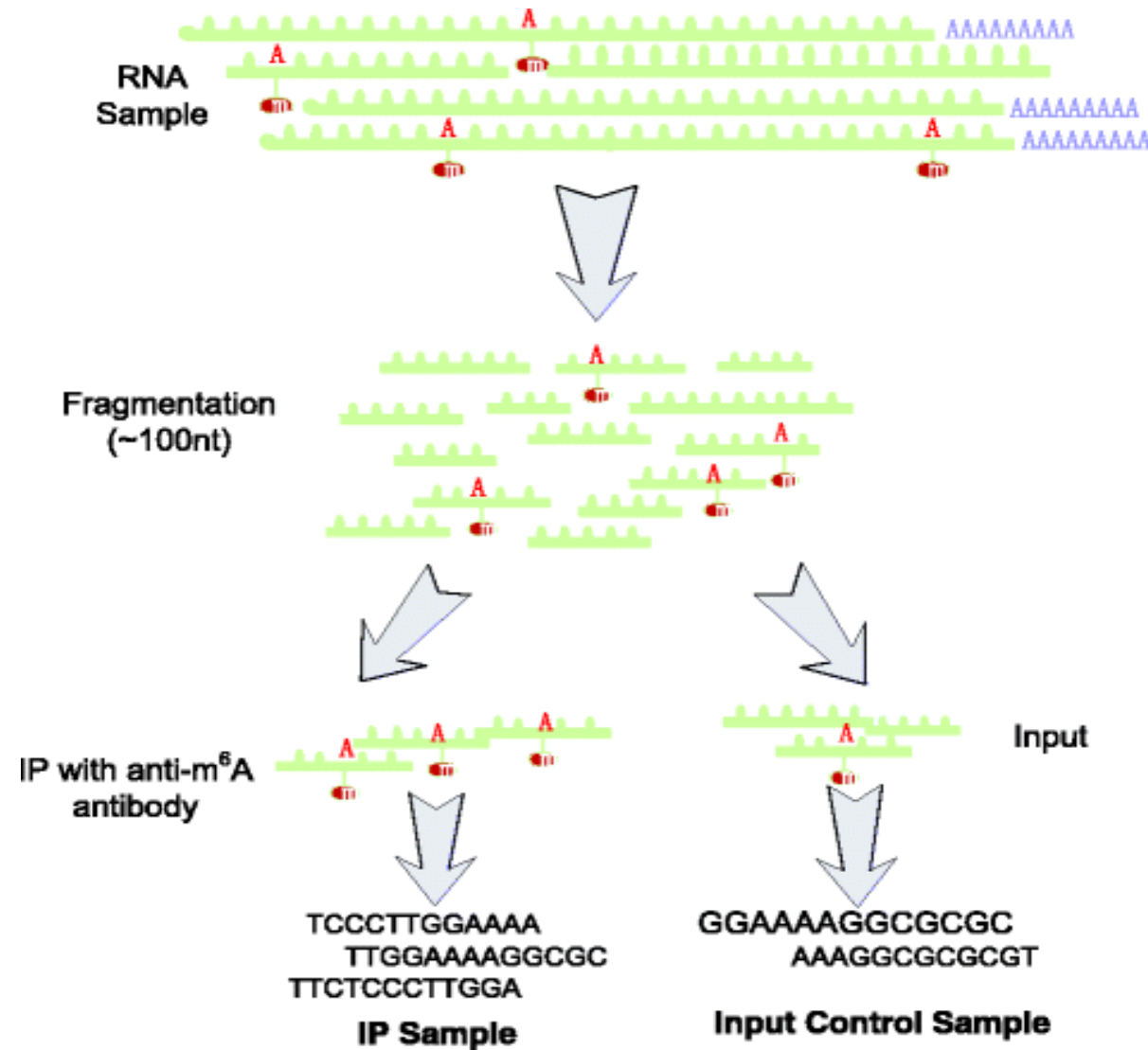
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4/14/2020

Organization of talk

- ❑ MeRIP protocol
- ❑ Coverage and number of replicates for good m6a signal
- ❑ Peak calling tools using in this paper
- ❑ Peak overlap between various studies
- ❑ Methodology used for differential peak methylation
- ❑ Evaluation of differential methylation peaks from various studies
- ❑ Tools for differential peak methylation
- ❑ Quad negative binominal – A review

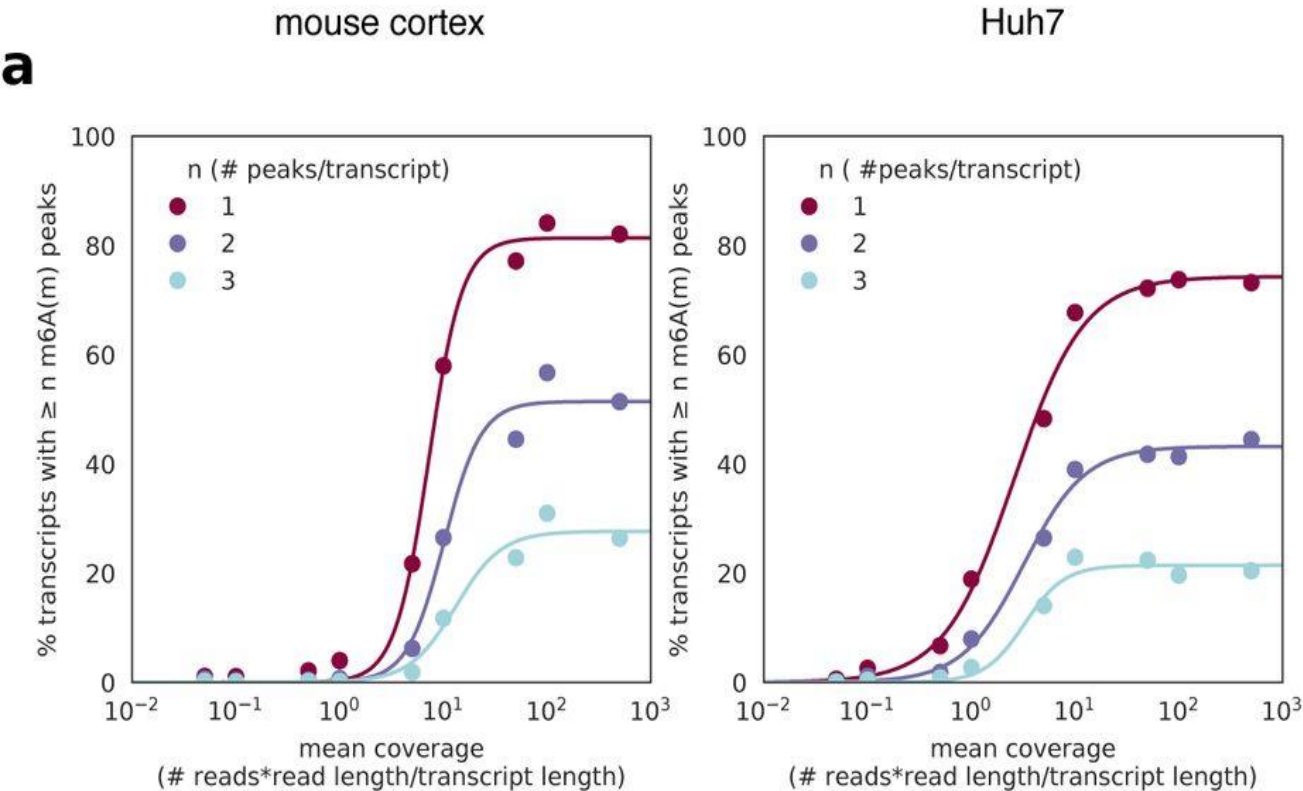
MeRIP protocol



IP Sample: measurement of methylated RNA fragments

Input samples: basal RNA abundance

Coverage of transcript and m6a peak site detection



Mouse Dataset :

Engel M, Eggert C, Kaplick PM, Eder M, Röh S, Tietze L, et al. The role of m6A/m-RNA methylation in stress response regulation. *Neuron*. 2018;99(2):389–403

Summary: Authors show that stress exposure and glucocorticoids region and time specifically alter m6A/m and its regulatory network. We demonstrate that deletion of the methyltransferase *Mettl3* or the demethylase *Fto* in adult neurons alters the m6A/m epitranscriptome, increases fear memory, and changes the transcriptome response to fear and synaptic plasticity

Samples : m6a of mouse cortex at baseline (7 replicates)

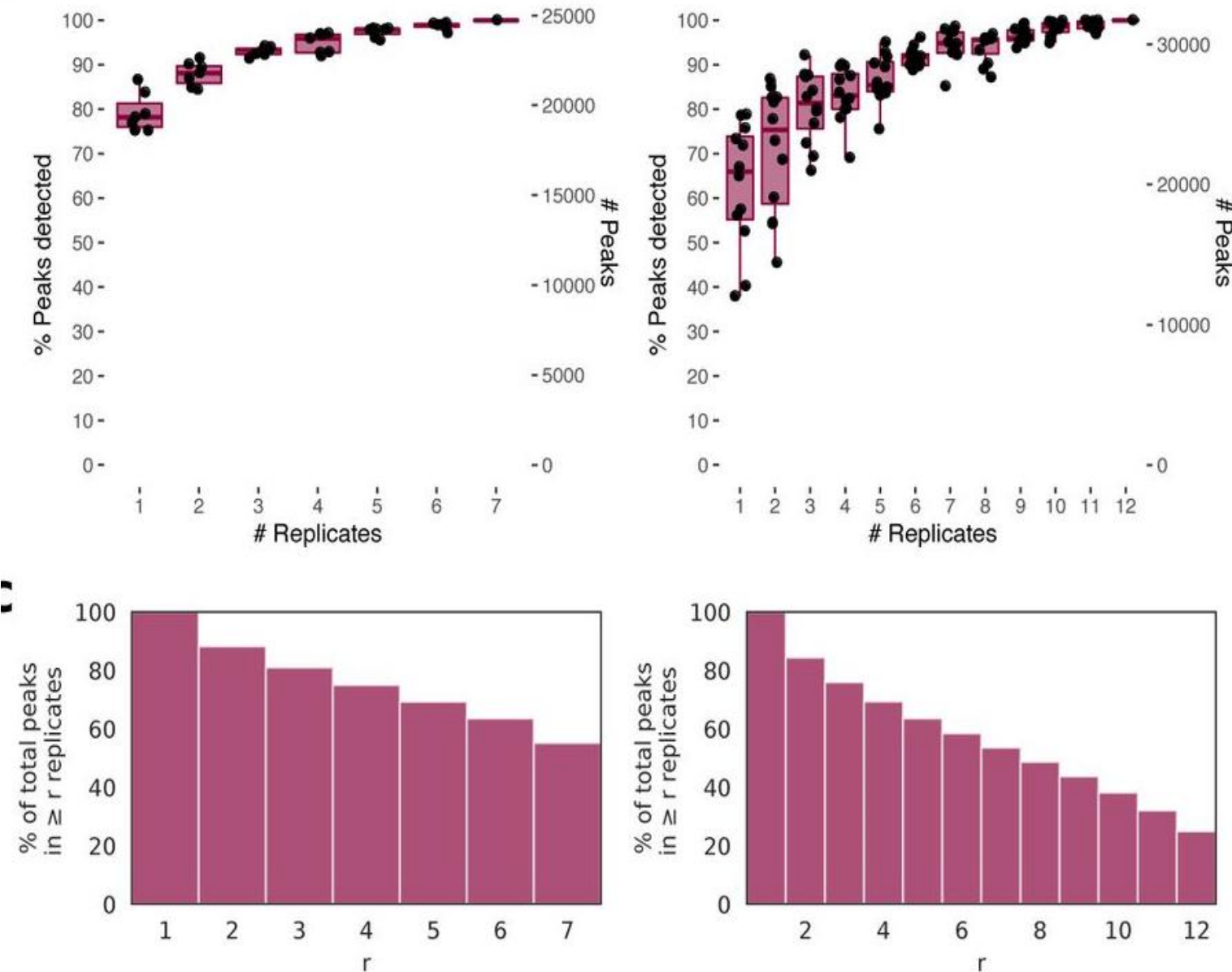
Human Dataset:

Gokhale NS, McIntyre ABR, Mattocks MD, Holley CL, Lazear HM, Mason CE, et al. Altered m6A modification of specific cellular transcripts affects Flaviviridae infection. *bioRxiv*. 2019 Jan

Summary: The authors analyzed m6A across the transcriptome following infection by dengue virus (DENV), Zika virus (ZIKV), West Nile virus (WNV), and hepatitis C virus (HCV). We found that infection by these viruses in the Flaviviridae family alters m6A modification of specific cellular transcripts, including *RIOK3* and *CIRBP*

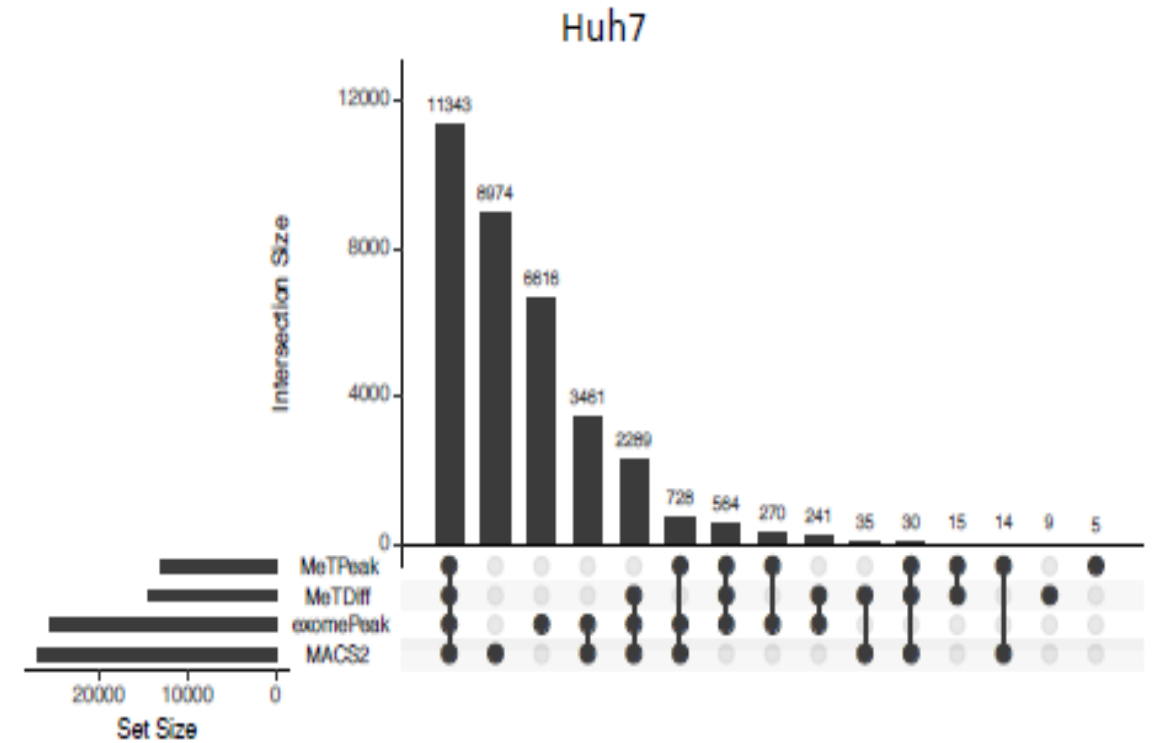
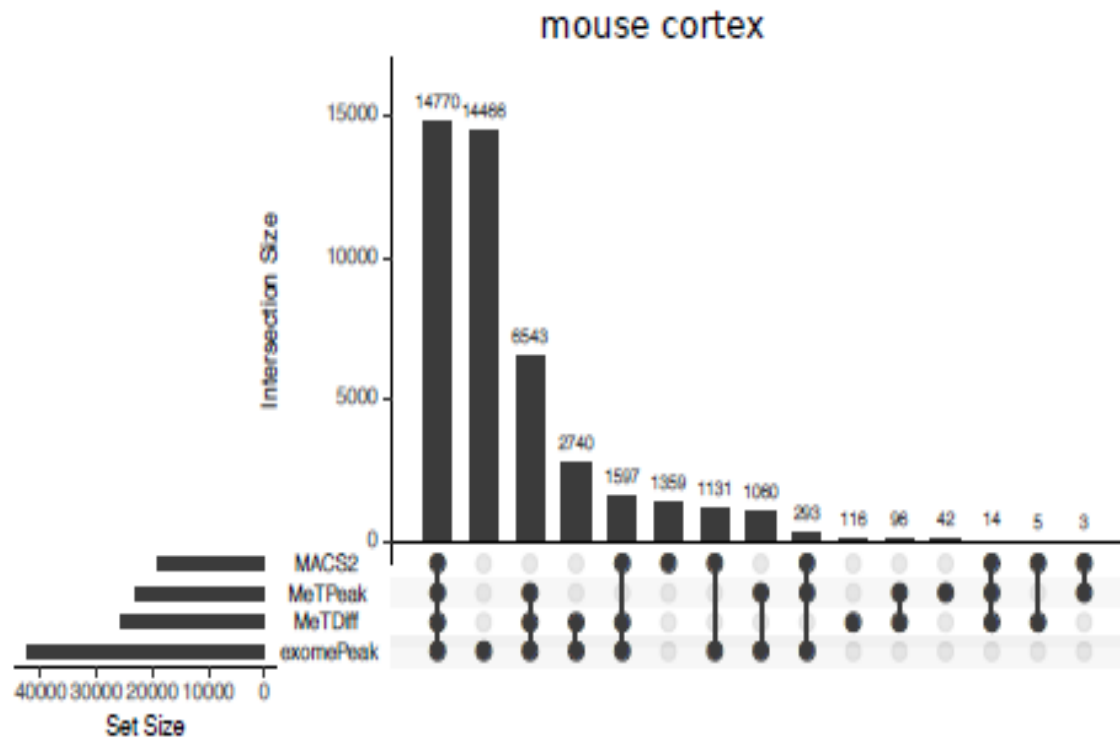
Samples : 12 replicates from uninfected human liver cells

Evaluation of number of replicates for good m6a peak sites



The total number of peaks captured increases with more replicates, with single replicates capturing a median of 66-78% of total peaks depending on study. Boxes span the 1st to 3rd quartiles of distributions for random subsamples of replicates, with lines indicating the median number of peaks, and whiskers showing the minimum and maximum points within ± 1.5 x the interquartile distance from the boxes. Jittered points show results for each random subsample (a total of 6 subsamples per replicate number for the mouse cortex data and 12 for the Huh7 data). **c)** The percent of peaks detected in at least r replicates for the same data sets

Peak detection tool used in the paper

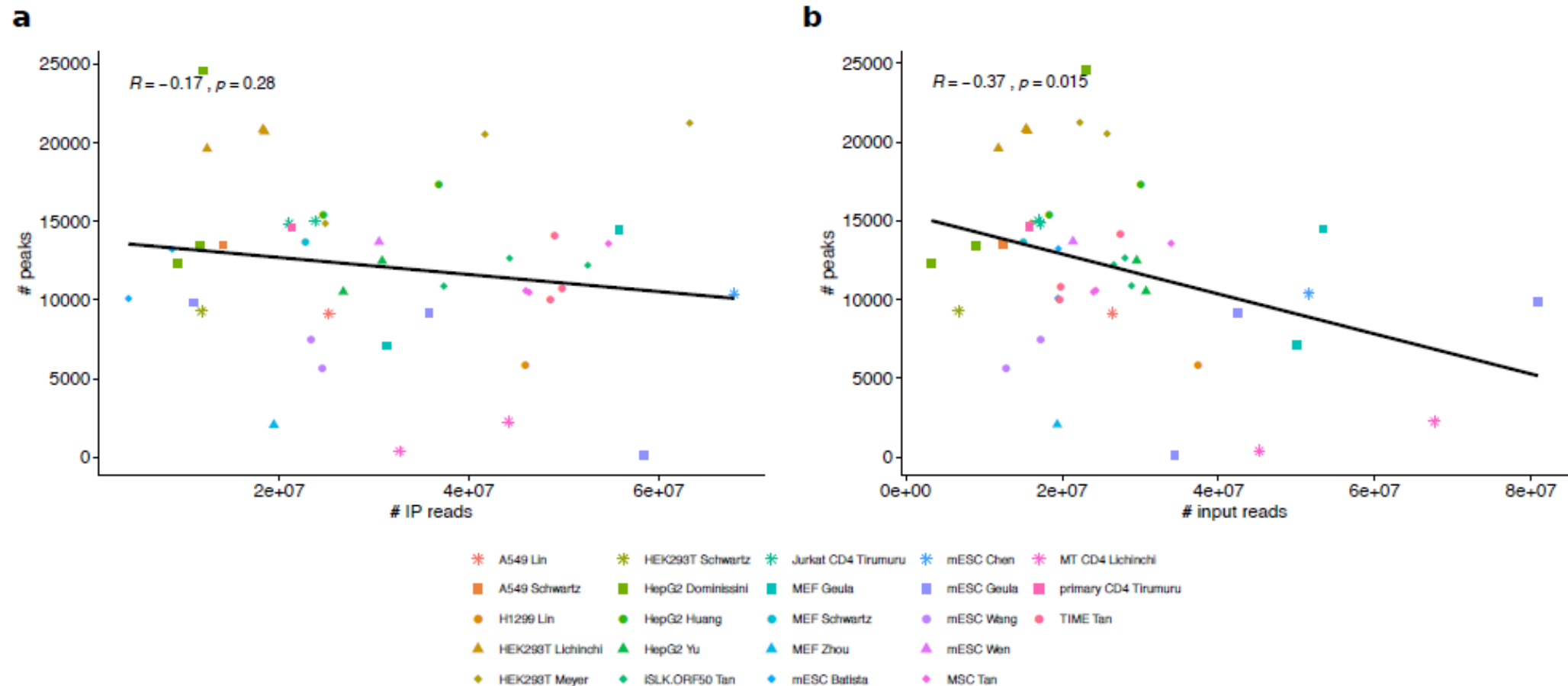


Conclusions till now

- For m6a peak detection a transcript must be efficiently expressed with coverage of 10X -50X
- Likely hood of false positive rises with lower coverage ($< 10X$)
- A single sample captured a median of 78% of the peaks found in seven replicates of mouse cortex data and 66% of peaks found in twelve replicates of Huh7 cell data. The number of detected peaks increased log-linearly with the addition of more replicates, such that with three replicates, 84-92% of the peaks found with 7-12 replicates were detected
- As we increase the number of replicate the common peaks decreases. while $\sim 80\%$ of peaks were detected in at least two replicates, only $\sim 60\%$ were detected in six replicates for both data sets and $\sim 25\%$ in all twelve replicates of Huh7 cell data

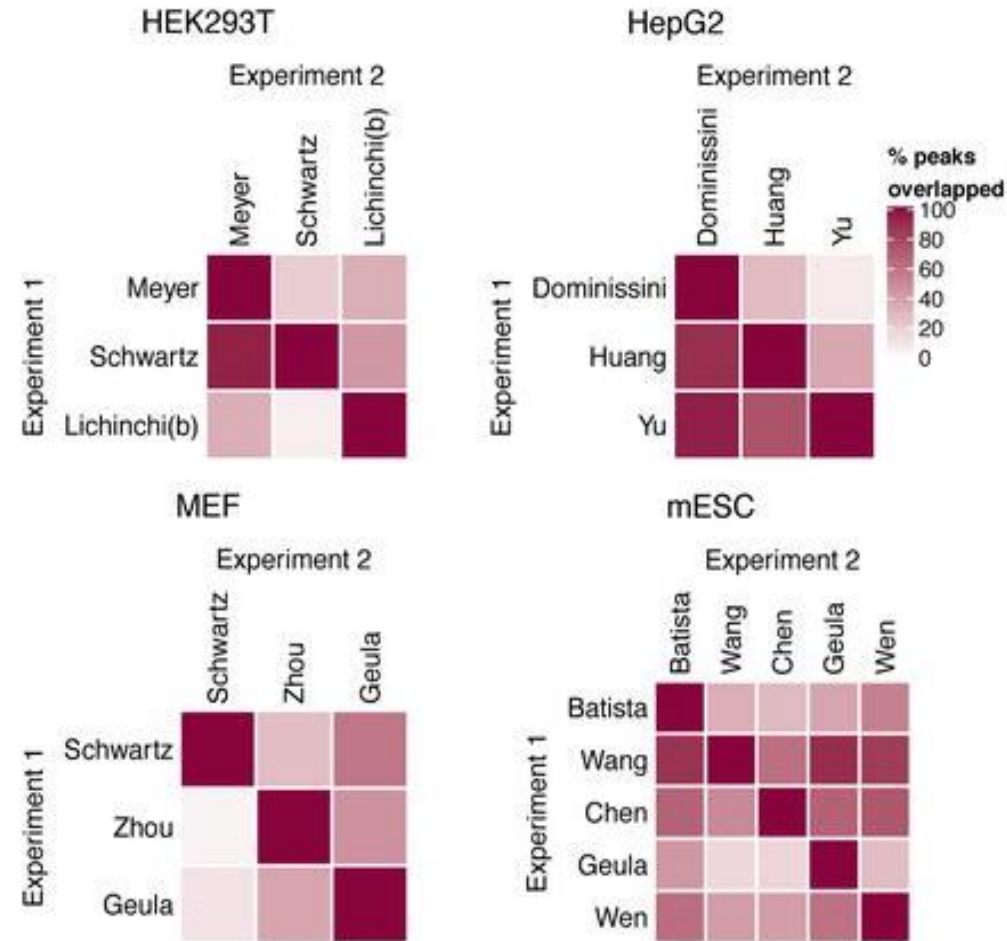
Correlation between sequencing depth and number of peaks

- Hypothesis is that variation in sequencing depth could contribute to the differences in peak counts.

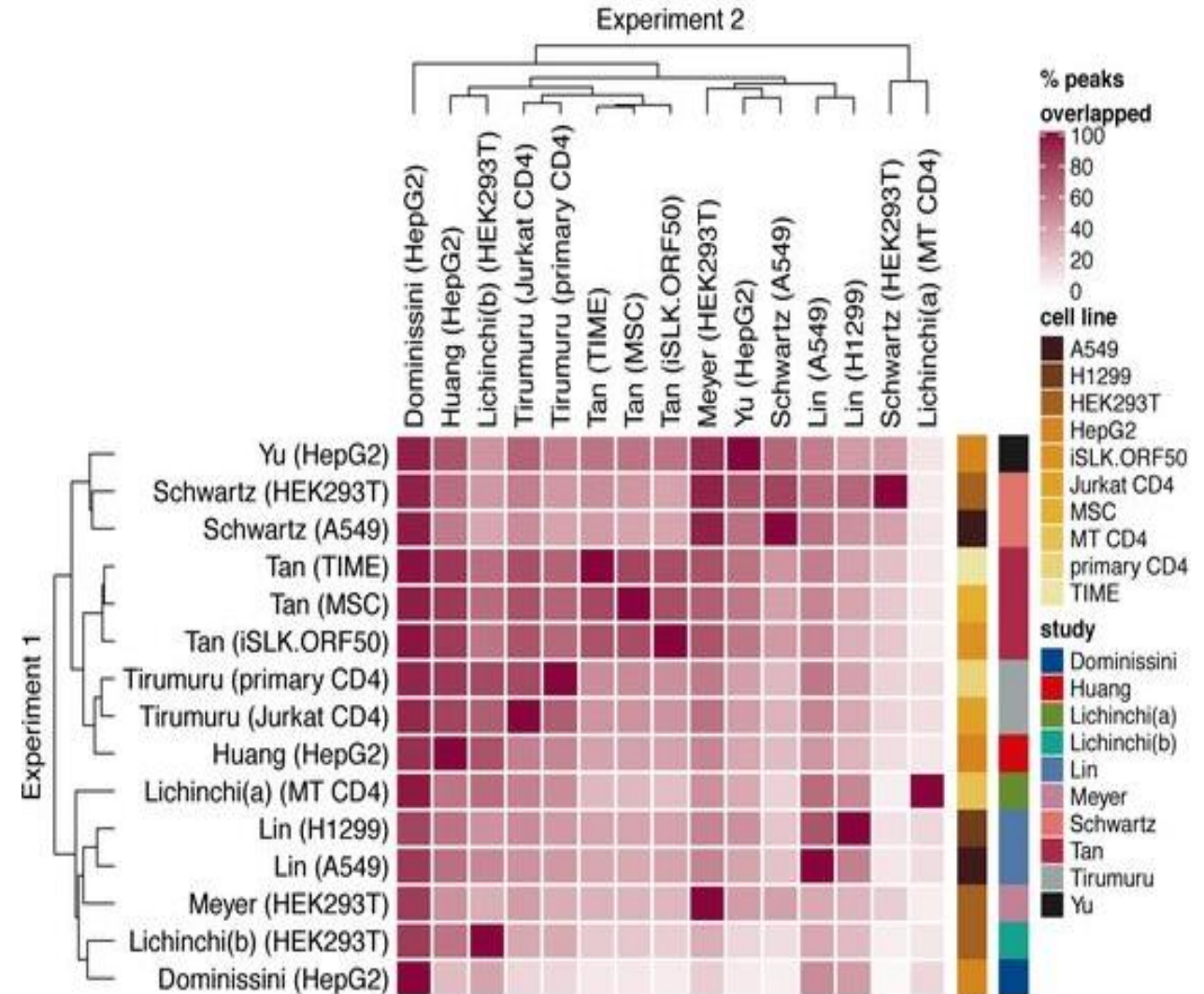


Peak overlap studies

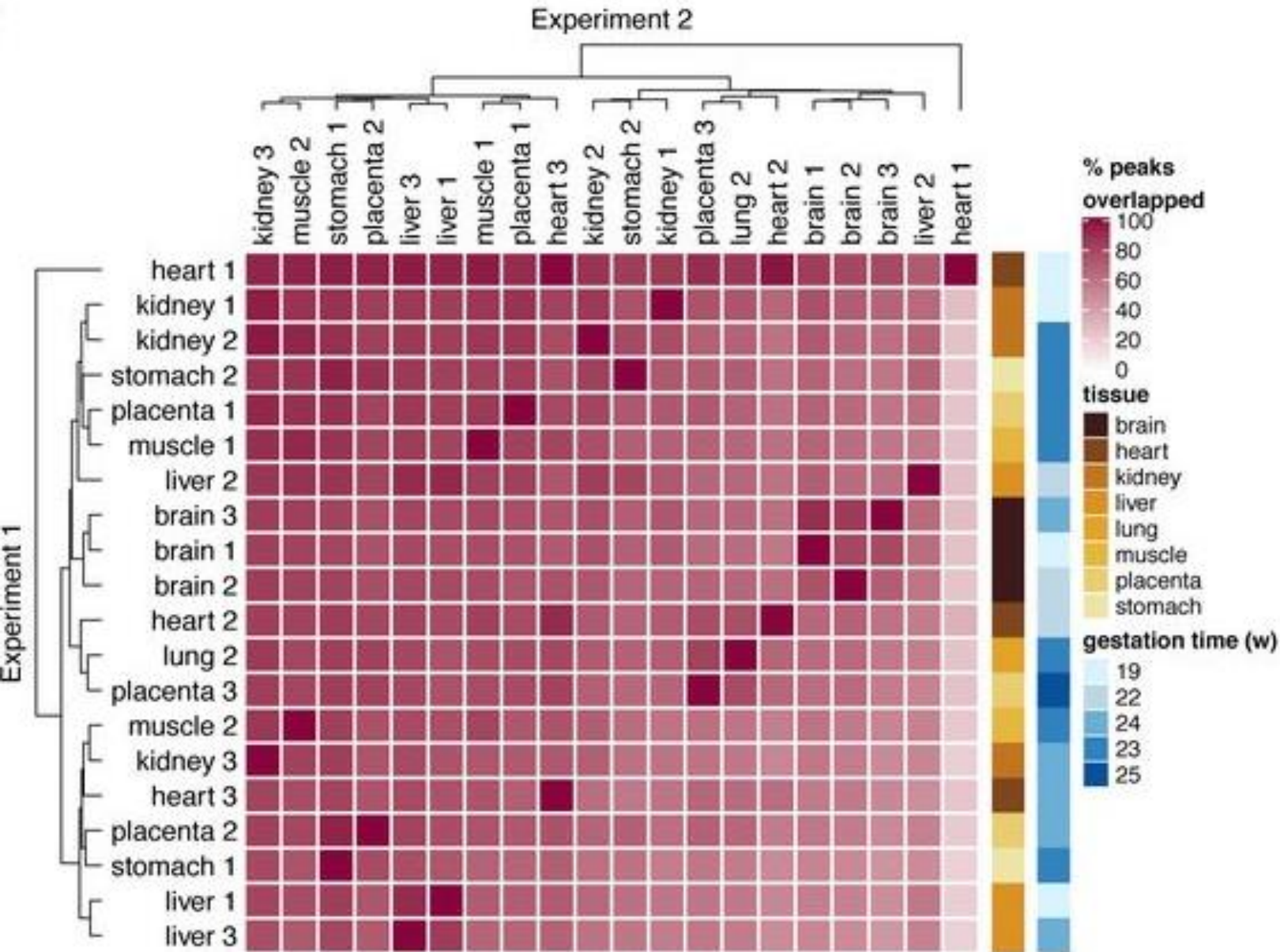
Peak overlap between same cell line from different studies



Peak overlap between different cell line within same studies shows more overlap in peaks



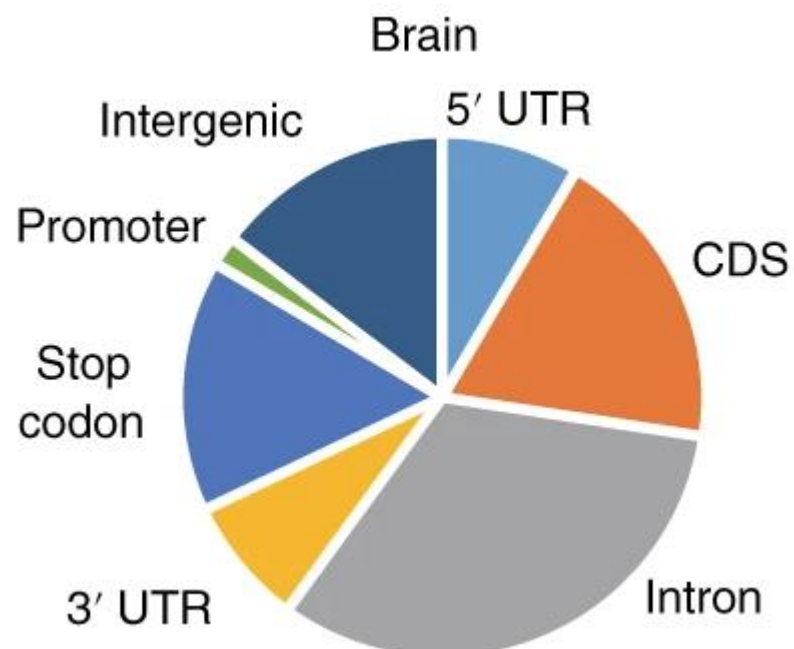
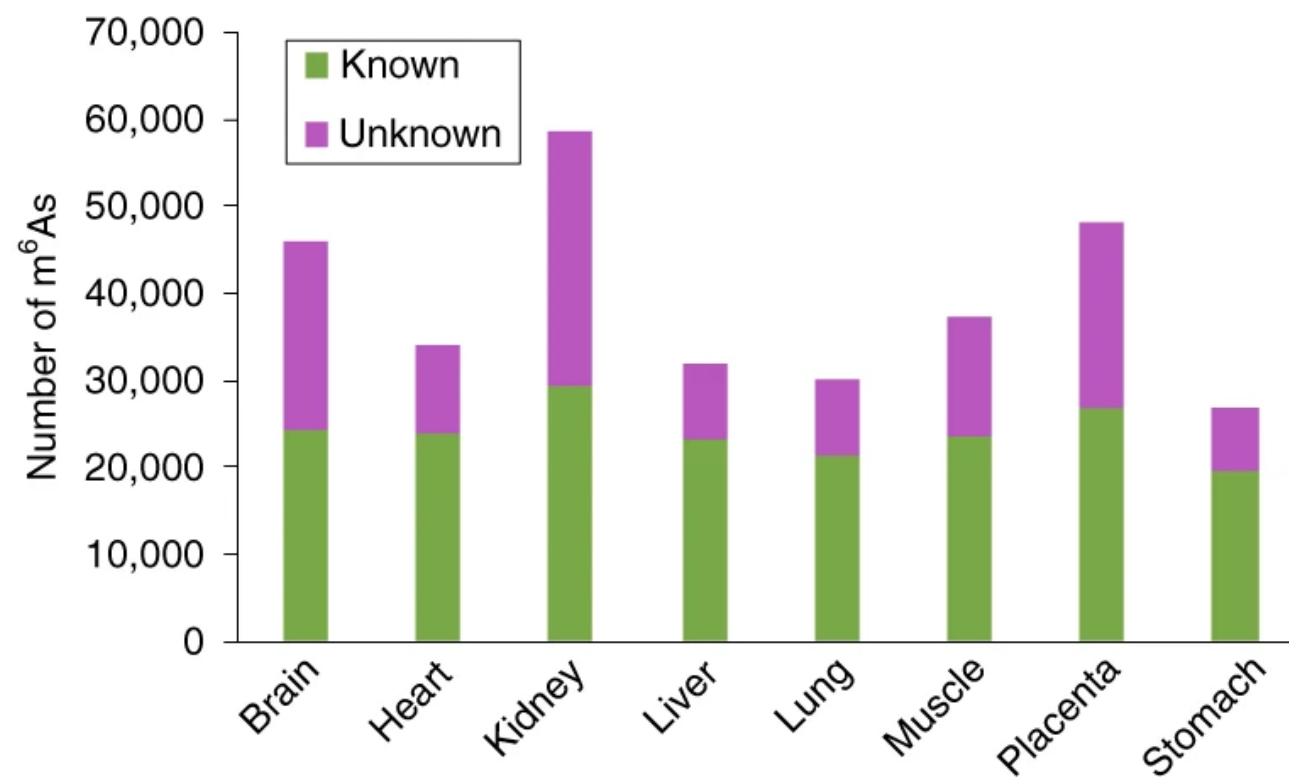
Peak detection across different tissue with a study



1) Eight major human fetal tissue were Collected from six fetuses in the second Trimester of pregnancy

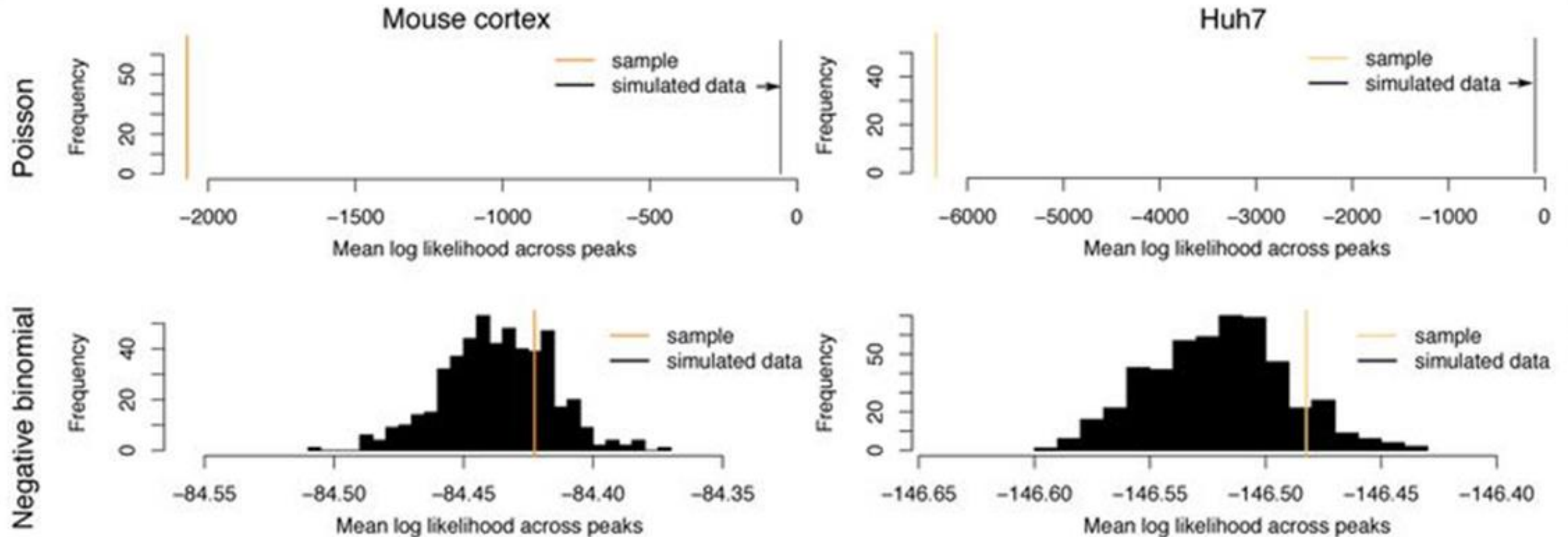
2) Peaks detected in different tissue analyzed In a single experiment showed high peak Overlap and little clustering by the tissue type

[Xiao S, Cao S, Huang Q, Xia L, Deng M, Yang M, et al. The RNA N6-methyladenosine modification landscape of human fetal tissues. Nature Cell Biology. 2019 May 1;21\(5\):651–61.](#)



Method and Tools

Negative binomial better models the distribution of read counts of m6A peaks

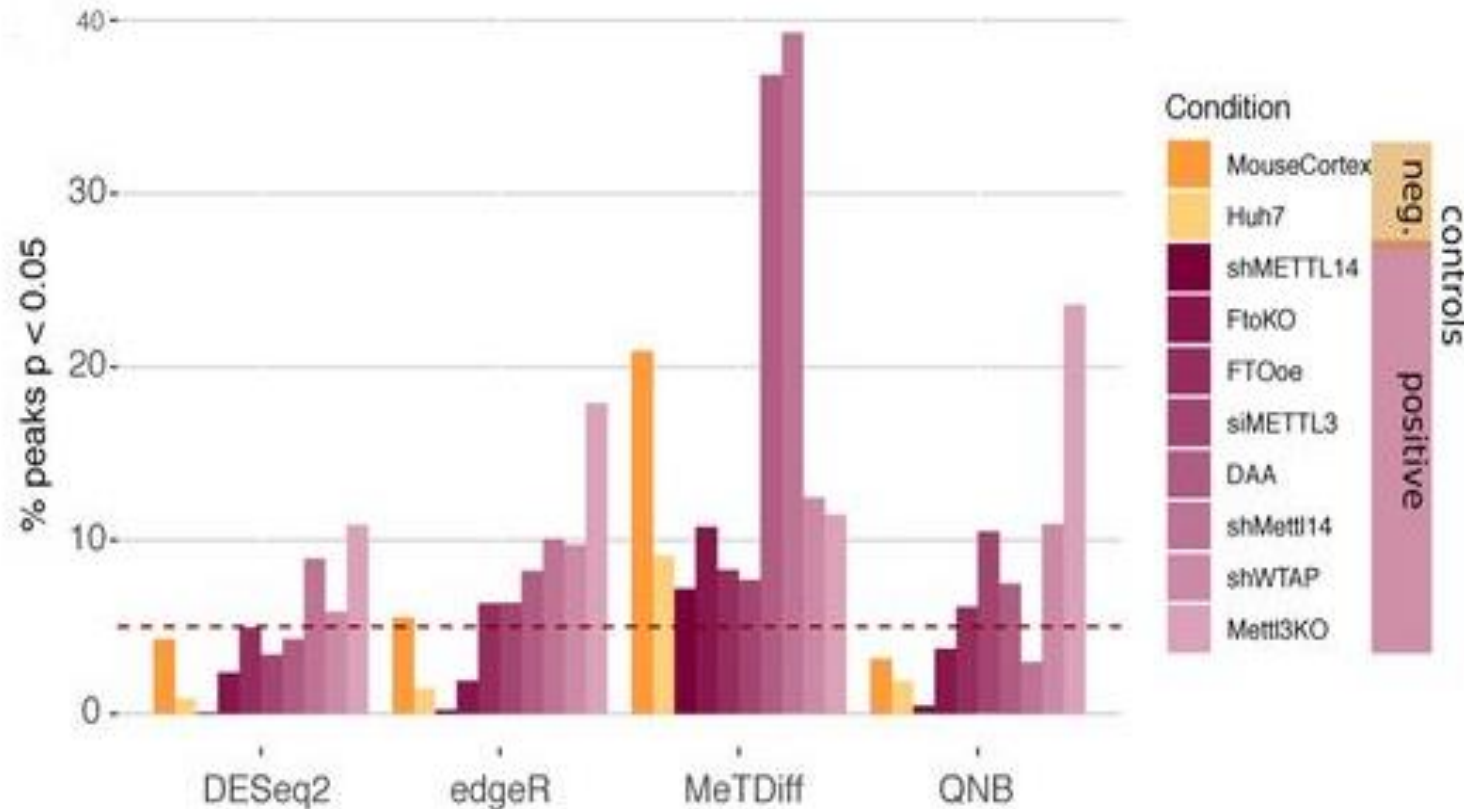


A comparison of Poisson (above) and negative binomial (below) models for read counts under peaks. The negative binomial mean log likelihood of the sample data fell within the 74th and 89st percentiles of 500 simulations for mouse cortex and Huh7 cell data, respectively, while the Poisson model failed to capture the sample distributions

Statistical methods for the detection of peak changes

Method	Read count distribution	Publication
MeTDiff	Poisson	Cui et al. (2018)
Quad-negative binomial (QNB)	Negative binomial	Liu et al. (2017)
GLM (DESeq2)	Negative binomial	based on Park et al. (2014) method for HITS-CLIP
GLM (edgeR)	Negative binomial	

Changes in peaks between two conditions in positive and negative controls

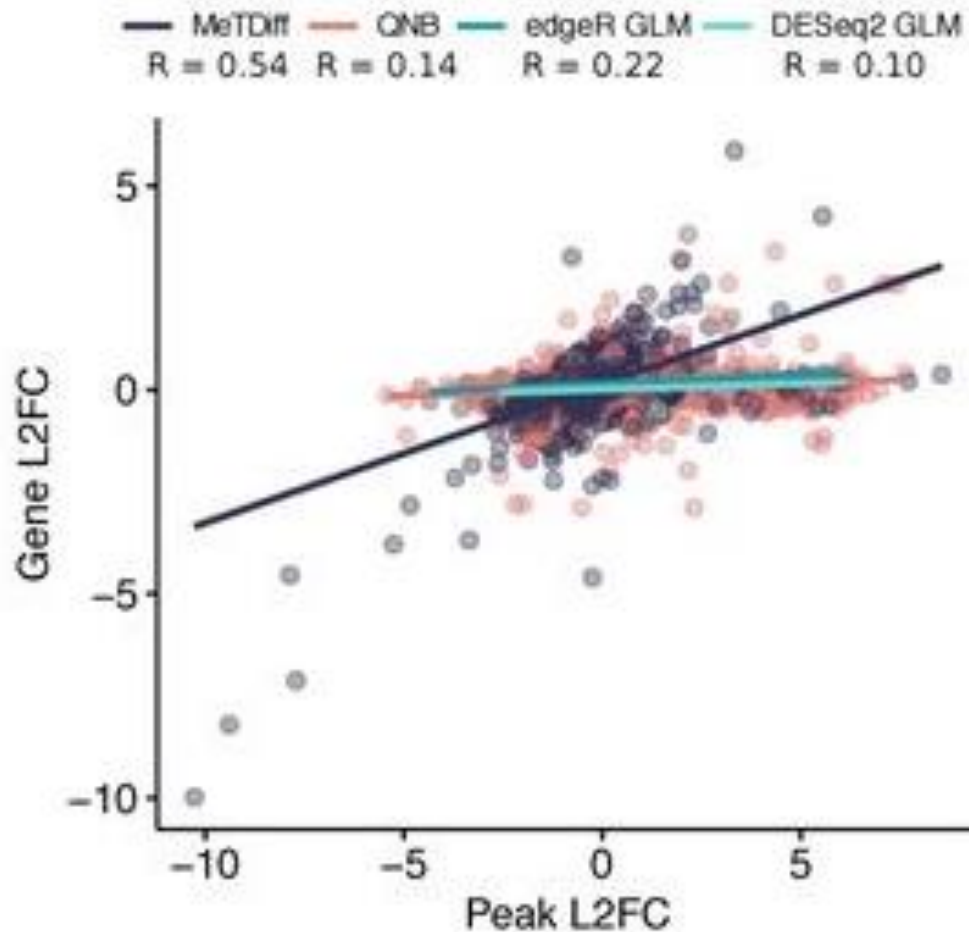


Negative control	Condition A	Condition B
Mouse (n=7) at baseline	3	4
Human (n =12) at baseline	6	6

Positive control tested the change in methylated peaks between wild type and knockout of methylation machinery in different studies of same cell line

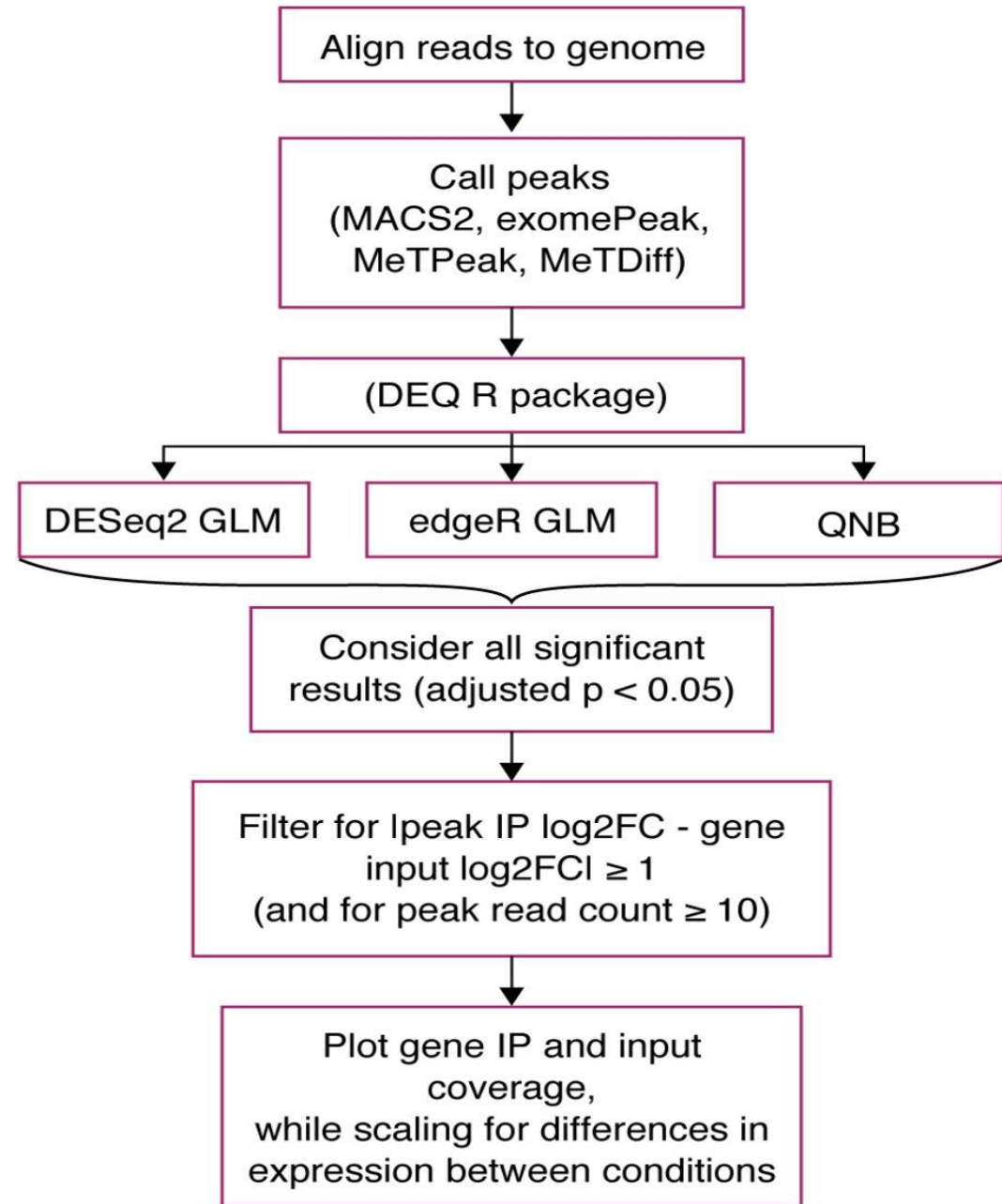
The line at 5% indicates the expected proportion of sites given a uniform p-value distribution

Correlation between changes in IP read counts at peak sites and changes in input read counts across their encompassing genes

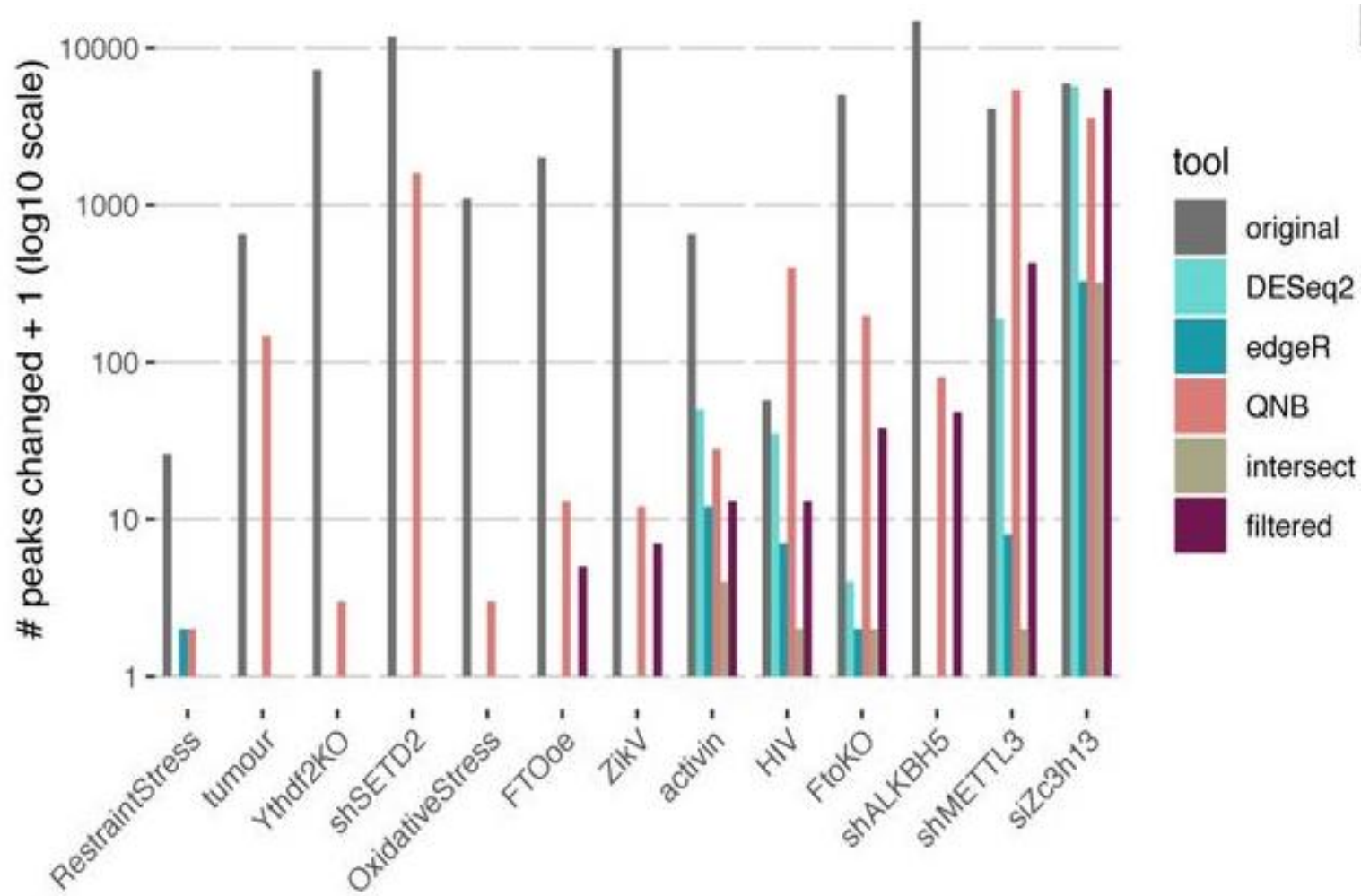


- ❑ The higher correlation for MeTDiff was driven by peaks with proportional changes in IP and input levels, which suggests that MeTDiff often detects differential expression of methylated genes rather than differential methylation
- ❑ **QNB or the GLM** implementations are better methods than MeTDiff to detect differential methylation

Analysis method developed in this paper



Reanalyzed peak changes between conditions



- 1) Fewer (0-100) significant differentially methylated peaks were reported
- 2) Minimum read threshold made a dramatic difference
- 3) In SETD2 dataset, out of 2065 sites identified by QNB, 2064 fell below the minimum read cut off

Quad –Negative binomial model

- Based on 4 NB distribution with mean and variance cross-linked to model IP and input control samples in 2 experimental condition
- Captures the within –group variability of RNA methylation level at small sample size
- Available as R package

QNB model

- When the sequencing depth of IP and input samples are equal, then

$$t_{i,j} \sim \text{Binomial}(p_{i,p(j)}, n_{i,j})$$

$t_{i,j}$ = read count for i feature in j^{th} replicate in IP sample

$c_{i,j}$ = read count for i feature in j^{th} replicate in input control sample

$$n_{i,j} = t_{i,j} + c_{i,j}$$

$P_{i,p(j)}$ = percentage of methylation in i feature in j^{th} replicate

$p(j)$ = experimental condition (cell type or tissue, treatment)

- Because of over-dispersion effect of sequencing data $t_{i,j}$ and $c_{i,j}$ follows NB distribution:

$$t_{i,j} \sim \text{NB}(\mu_{t,i,j}, \sigma_{t,i,j}^2)$$

$$c_{i,j} \sim \text{NB}(\mu_{c,i,j}, \sigma_{c,i,j}^2)$$

Where mean is

$$\mu_{t,i,j} = q_i p_{i,\rho(j)} e_{i,\rho(j)} s_{t,j}$$

$$\mu_{c,i,j} = q_i (1 - p_{i,\rho(j)}) e_{i,\rho(j)} s_{c,j}$$

Here, q_i represents the expected abundance of feature i under all conditions in a standard sequencing library. $s_{t,j}$ and $s_{c,j}$ represent the size factor of the IP and input control sample of the j -th biological replicate and directly reflect their sequencing depth. $p_{i,\rho(j)}$ stands for risk of RNA methylation, or the true percentage of methylation for feature i under condition $\rho(j)$ on the common scale, i.e., without rescaling by the size factors $s_{c,j}$ and $s_{t,j}$. Additionally, $e_{i,\rho(j)}$ is introduced to model differential expression at RNA level as a feature-specific size factor, which indicates the abundance of feature i under a specific experimental condition compared with the standard abundance q_i .

$t_{i,j}$ = read count for i feature in j^{th} replicate in IP sample

$c_{i,j}$ = read count for i feature in j^{th} replicate in input control sample

$$n_{i,j} = t_{i,j} + c_{i,j}$$

$P_{i,\rho(j)}$ = percentage of methylation in i feature in j^{th} replicate

$p(j)$ = experimental condition (cell type or tissue, treatment)

Calculation of different parameters

$$\hat{q}_i = \mathbb{E}_{\forall j} \left(\frac{t_{i,j}}{s_{t,j}} + \frac{c_{i,j}}{s_{c,j}} \right)$$

--equation 1

$$\hat{p}_{i,\rho(j)} = \sum_{j:\rho(j)=\rho} \left(\frac{t_{i,j}}{s_{t,j}} \right) / \sum_{j:\rho(j)=\rho} \left(\frac{t_{i,j}}{s_{t,j}} + \frac{c_{i,j}}{s_{c,j}} \right)$$

--equation 2

$$\hat{e}_{i,\rho} = \frac{1}{|\rho| \hat{q}_i} \sum_{j:\rho(j)=\rho} \left(\frac{t_{i,j}}{s_{t,j}} + \frac{c_{i,j}}{s_{c,j}} \right)$$

--equation 3

$|\rho|$ = number of biological replicates

Decomposing variance

$$\sigma_{t,i,j}^2 = \underbrace{\mu_{t,i,j}}_{\text{shot noise}} + \underbrace{(e_{i,j}s_{t,j})^2 v_{t,i,\rho(j)}}_{\text{raw variance}}$$

$$\sigma_{c,i,j}^2 = \underbrace{\mu_{c,i,j}}_{\text{shot noise}} + \underbrace{(e_{i,j}s_{c,j})^2 v_{c,i,\rho(j)}}_{\text{raw variance}}$$

where $\mu_{t,i,j}$ and $\mu_{c,i,j}$ are the variance of a Poisson distribution, which is often used to model technical replicates in NGS data. Additionally, due to biological variability, the over-dispersion of a Poisson model is represented by $(e_{i,\rho(j)}s_{t,j})^2 v_{t,i,\rho(j)}$ and $(e_{i,\rho(j)}s_{c,j})^2 v_{c,i,\rho(j)}$, where $e_{i,\rho(j)}$ and $s_{t,j}$ (or $s_{c,j}$) quantify the impact of condition-specific gene differential expression and sample-specific library size (or the sequencing depth), respectively. We consider the per-feature raw variance parameter $v_{i,\rho}$ is a smooth function of the expected methylation rate $p_{i,\rho}$ and the feature abundance $q_{i,\rho}$ under a specific condition ρ , i.e.,

$$v_{t,i,\rho(j)} = v_{t,\rho}(p_{i,\rho(j)}, q_{i,\rho(j)}) \quad \text{--equation 4}$$

$$v_{c,i,\rho(j)} = v_{c,\rho}(p_{i,\rho(j)}, q_{i,\rho(j)}) \quad \text{--equation 5}$$

For methylation reads count $t_{i,j}$ in the IP sample, the variances on the common scale $\hat{w}_{t,i,\rho}$ can be calculated with

$$\hat{w}_{t,i,\rho} = \frac{1}{(|\rho|-1)} \sum_{j:\rho(j)=\rho} \left[\frac{t_{i,j}}{\hat{s}_{t,j}\hat{e}_{i,\rho(j)}} - \bar{q}_{t,i,\rho} \right]^2 \quad \text{--equation 6}$$

where

$$\bar{q}_{t,i,\rho} = \frac{1}{|\rho|} \sum_{j:\rho(j)=\rho} \frac{t_{i,j}}{\hat{s}_{t,j}\hat{e}_{i,\rho(j)}} \quad \text{--equation 7}$$

Let

$$z_{t,i,\rho} = \frac{\hat{q}_i \hat{p}_{i,\rho(j)}}{|\rho|} \sum_{j:\rho(j)=\rho} \left(\frac{1}{\hat{s}_{t,j}\hat{e}_{i,\rho(j)}} \right) \quad \text{--equation 8}$$

$\left(\hat{w}_{t,i,\rho} - z_{t,i,\rho} \right) =$ unbiased estimator of raw variance parameter $\boldsymbol{v}_{t,i,\rho}$

$$\hat{v}_{t,i,\rho(j)} \left(\hat{p}_{i,\rho}, \hat{q}_i \right) = w_{t,i,\rho} \left(\hat{p}_{i,\rho}, \hat{q}_i \right) - z_{t,i,\rho} \quad \text{--equation 9}$$

- 2 dimensional local regression on graph $(\hat{p}_{i,\rho}, \hat{q}_i, \hat{w}_{t,i,\rho})$ is used to obtain a smooth function of $w_{t,i,\rho}(\hat{p}_{i,\rho}, \hat{q}_i)$.
- Also implemented a GLM of gamma family for local regression with the implementation in R locfit package for estimation of $w_{t,i,\rho}(\hat{p}_{i,\rho}, \hat{q}_i)$.
- Similarly the variance of read count for input control sample is calculated.

Testing and metrics

- Null hypothesis : there are no changes in methylation rate in condition A and B on the common scale i.e., $p_{i,A} = p_{i,B} = p_{i,O}$

$$\hat{p}_{i,O} = \sum_{j \in A \cup B} \frac{t_{i,j}}{s_{t,j}} / \sum_{j \in A \cup B} \left(\frac{t_{i,j}}{s_{t,j}} + \frac{c_{i,j}}{s_{c,j}} \right)$$

- For differential methylation analysis between condition A and B, 4 different negative binomial models are constructed for IP and input control samples under two experimental condition i.e ;

$$t_{i,A} = \sum_{j \in A} (t_{i,j}) \sim \text{NB}(\hat{\mu}_{t,i,A}, \hat{\sigma}_{t,i,A}^2)$$

$$t_{i,B} = \sum_{j \in B} (t_{i,j}) \sim \text{NB}(\hat{\mu}_{t,i,B}, \hat{\sigma}_{t,i,B}^2)$$

$$c_{i,A} = \sum_{j \in A} (c_{i,j}) \sim \text{NB}(\hat{\mu}_{c,i,A}, \hat{\sigma}_{c,i,A}^2)$$

$$c_{i,B} = \sum_{j \in B} (c_{i,j}) \sim \text{NB}(\hat{\mu}_{c,i,B}, \hat{\sigma}_{c,i,B}^2)$$

- Distribution parameters for each negative binomial model is calculated. For example;

$$\hat{\mu}_{t,i,A} = \hat{p}_{i,O} \hat{q}_i \hat{e}_{i,A} \sum_{j \in A} \hat{s}_{t,j}$$

$$\hat{\sigma}_{t,i,A}^2 = \hat{p}_{i,O} \hat{q}_i \hat{e}_{i,A} \sum_{j \in A} s_{t,j} + v_A \left(\hat{p}_{i,O}, \hat{q}_i \right) \hat{e}_{i,A}^2 \sum_{j \in A} \hat{s}_{t,j}^2$$

Given the total number of methylation read count ($t_i = t_{i,A} + t_{i,B}$) and the total number of reads under each condition ($n_{i,A} = t_{i,A} + c_{i,A}$) and ($n_{i,B} = t_{i,B} + c_{i,B}$) do not change, the joint conditional probability of the observation ($t_{i,A} = t$) can be calculated with

$$P(t_{i,A} = t | t_i, n_{i,A}, n_{i,B}) = P(t_{i,A} = t) P(t_{i,B} = t_i - t) \\ P(c_{i,A} = n_{i,A} - t) P(c_{i,B} = n_{i,B} - t_i + t)$$

- Statistical significance is calculated as

$$p\text{-value} = \frac{\sum_{t:P(t) \leq P(t_{i,A})} P(t)}{\sum_{\forall t} P(t)}$$

- The degree of differential methylation is given by ;

$$RR_i = \hat{p}_{i,A} / \hat{p}_{i,B}$$

Where A is treatment group and B is control group

How this tool is different from other

- Over-dispersion in the input control samples are also modeled in QNB. In other tools, input control samples are used for gene expression estimation
- The inclusion of input control samples in the test, rather than simply using it in background boosts the performance of QNB.

Other features of QNB

- QNB allows different modes of running the tool
 - a) Per –condition
 - b) Pooled : calculate a single dispersion value using samples in all the conditions with replicates
 - c) Blind : ignores sample labels and estimates the dispersion value as if all the samples were replicates for a single condition
 - d) Auto : selects per-condition mode if biological replicates are present else selects blind mode.

```
.sizeFactor2 <-
function(cbind(control_ip,treated_ip,control_input,treated_input)) {
  temp <- log(colSums(n))
  temp <- temp - mean(temp)
  s_size <- exp(temp)
  return(s_size)
}
```

Size factor function

```
function(meth1,meth2,unmeth1,unmeth2,s_t1,s_t2,s_c1,s_c2){
#estimate probability of methylation under a condition
p1 <- .estimateP(meth1, unmeth1, s_t1, s_c1)
p2 <- .estimateP(meth2, unmeth2, s_t2, s_c2)
p0 <- .estimateP(cbind(meth1,meth2), cbind(unmeth1,unmeth2),
c(s_t1,s_t2), c(s_c1,s_c2))
```

Function for calculation of P for each condition

```
.estimateP <- function(meth, unmeth, size_t, size_c) {
  temp_t <- t(t(meth)/size_t)
  temp_c <- t(t(unmeth)/size_c)
  temp_n <- temp_t+temp_c
  p <- rowSums(temp_t)/rowSums(temp_n)
  p[is.na(p)] <- 0.5
  p[is.infinite(p)] <- 0.5
  return(p)
}
```



```
.estimateQ <- function(meth,unmeth,size_t,size_c,p,useAll=TRUE){
  if (useAll) {
    temp_t <- t(t(meth)/size_t)
    temp_c <- t(t(unmeth)/size_c)
    temp_n <- temp_t+temp_c
    q <- rowSums(temp_n)/length(size_t)
  } else {
    temp_c <- t(t(unmeth)/size_c)
    qc <- rowMeans(temp_c)
    q <- qc/(1-p)
  }

  q[is.na(q)] <- 0
  return(q)
}
```

Function for calculation of Q for each condition

```
.estimateE <- function(meth,unmeth,size_t,size_c,q){
  temp_t <- t(t(meth)/size_t)
  temp_c <- t(t(unmeth)/size_c)
  temp_n <- temp_t+temp_c
  e <- rowSums(temp_n)/length(size_t)/q
}
```

Function for calculation of E for each condition

```
.calculateW <- function(meth,size_t,e_t){  
  temp_t <- t(t(meth)/size_t)  
  q <- temp_t/e_t  
  # q[is.na(q)] <- 0  
  resi <- q-rowMeans(q)  
  w <- rowSums(resi^2)/(length(size_t)-1)  
  w <- pmax(w,1e-8)  
  return(w)  
}  
~
```

```
.fittedW <- function(p,q,fit){  
  l <- log(q+1)  
  c <- c(rep(1,length(p)))  
  data=data.frame(cbind(p,l,c))  
  w_fit <- predict(fit,data)  
  return(w_fit)  
}  
~
```